

МОСКОВСКИЙ ФИЗИКО-ТЕХНИЧЕСКИЙ ИНСТИТУТ
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ESTIMATION IN NEURAL ENSEMBLE SEARCH**

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Annotation

Ensembles are a standard way to improve the performance and robustness of deep neural networks, but their effectiveness crucially depends on both the quality and the diversity of individual models. Most neural architecture search (NAS) methods are computationally expensive. Extending them to neural ensemble search (NES), which requires joint optimization of individual architectures and their ensemble composition, leads to an exponential growth of the search space and makes the problem computationally intractable. To address this, we introduce a dual-objective surrogate-guided ensemble search: candidate architectures are represented as directed acyclic graphs, and two surrogate models are trained independently to estimate predictive accuracy and diversity potential. Their combined estimates guide an NES framework that efficiently identifies architectures that are both individually strong and collectively diverse. Our final ensemble achieves competitive or superior performance compared to standard baselines such as Deep Ensembles and Random Search on FashionMNIST, CIFAR-10, and CIFAR-100.

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Introduction

Neural network ensembles often demonstrate better accuracy, improved robustness, and more reliable uncertainty estimation compared to single models, especially in classification and regression tasks [1, 2]. This motivates the development of methods for efficient ensemble construction, formalized as Neural Ensemble Search (NES) [3, 4]. NES, in turn, relies on Neural Architecture Search (NAS) methods, which are extensively studied and applied to search for individual neural network architectures, such as evolutionary algorithms [5, 6], reinforcement learning [7, 8], and Bayesian optimization [9, 10]. Selecting an optimal architecture for even a single model is a challenging task, particularly when considering data-specific constraints and computational limitations [11, 12].

One of the earliest works for NES is DeepEns [13], which trains the same neural network architecture multiple times with different random initializations and combines their predictions. While this method is simple to implement and tune, its effectiveness is limited because there is no guarantee that the resulting ensemble will be sufficiently diverse. More sophisticated adaptations of NAS techniques are presented in recent works [4, 14, 15], which are designed to efficiently combine multiple networks into an ensemble. However, these methods often lack explicit, structured control over ensemble diversity: diversity either emerges implicitly through optimization objectives [4] or is guided by heuristic similarity measures [15], without a dedicated mechanism to balance it against individual model accuracy.

Our research adapts ideas from NAS for NES, specifically utilizing surrogate functions to avoid the expensive evaluation of numerous candidate architectures [16, 17]. Modern NAS methods extensively employ surrogate models to estimate architecture performance without requiring full training, which significantly reduces computational overhead [16, 18]. For instance, the work of Lu et al. [17] demonstrates the effectiveness of surrogate-assisted evolutionary algorithms in multi-objective search spaces. In the context of specialized tasks, the study by Calisto et al. [19] utilizes a Surrogate-assisted Multiobjective Evolutionary-based Algorithm (SaMEA) for medical image segmentation, demonstrating the robustness of surrogates in high-dimensional domains. Furthermore, recent

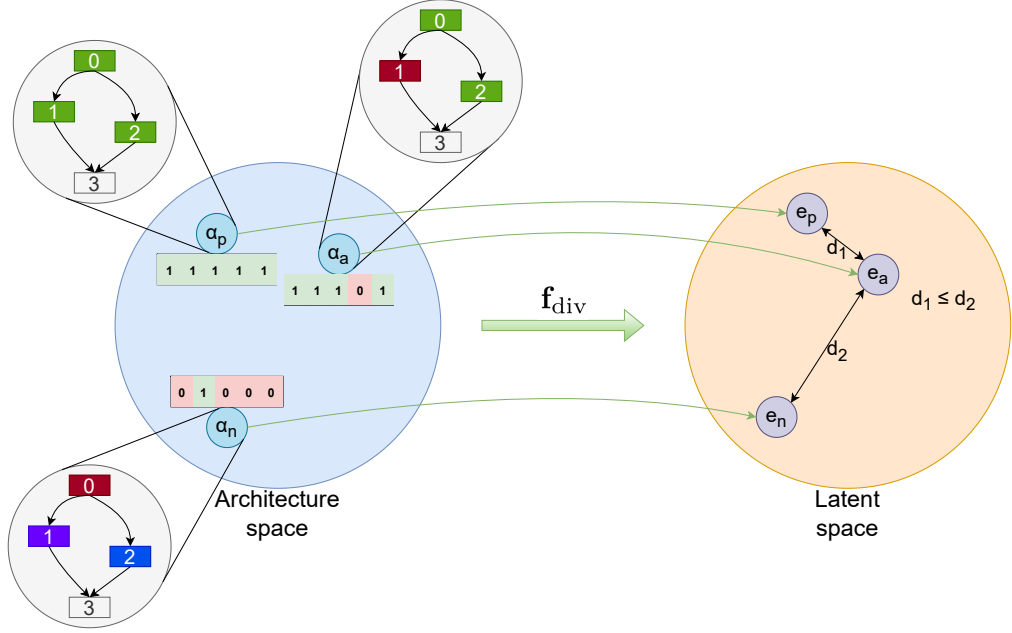


Figure 1 — Visualization of the surrogate diversity function. Architecture space showing different architectures of neural networks represented as graphs, the vectors shown nearby each architecture represent the predictions of these models on a fixed dataset. A surrogate diversity function transforms this space into latent space. In latent space where similar architectures (e.g. $\mathbf{e}_a$ and $\mathbf{e}_p$) are mapped close together, while dissimilar architectures (e.g., $\mathbf{e}_a$ and $\mathbf{e}_n$) are mapped farther apart, satisfying $d_1 \leq d_2$. Geometric organization in a latent space makes it possible to effectively evaluate diversity without the need for pre-training models with these architectures.

developments in similarity-based surrogates [20] provide a foundation for using such models to estimate not only accuracy but also structural relationships between architectures.

To address the challenges of NES, we propose a dual-objective surrogate-guided ensemble framework. This framework separates the search into two predictive objectives: accuracy estimation for individual architectures and *diversity* modeling via prediction of architecture embeddings in a latent space [20], enabling diversity assessment through embedding-space geometry. Inspired by surrogate-assisted NAS [12, 19], we encode candidate architectures as graphs

and train lightweight surrogate models to capture these objectives. As illustrated in Fig. 1, the surrogate diversity function maps architectures from the discrete search space into a continuous latent embedding. In this space, architectures with correlated predictions are embedded close to each other, while dissimilar ones are separated by larger distances. This geometric organization enables efficient, gradient-friendly, and explicit estimation of ensemble diversity without requiring costly model training. To implement this, we employ Graph Attention Networks (GATs) [21] as the backbone for surrogate modeling [16]. We utilize a Triplet Loss objective [22] to structure the latent space according to architectural diversity.

Main Contributions:

1. We introduce a novel surrogate-based methodology for training diversity predictors that map discrete neural architectures to a structured latent space.
2. We present the surrogate-assisted framework for Neural Ensemble Search that jointly optimizes predictive performance and architectural diversity via dual-objective guidance. To the best of our knowledge, this is one of the first methods to explicitly model diversity in NES using surrogate-based latent representations.
3. We provide a proof-of-concept validation of our framework on FashionMNIST [23], CIFAR-10 [24], and CIFAR-100 [24], demonstrating its capability to balance predictive performance and architectural diversity effectively.
4. We provide datasets of approximately 3,000 trained models per dataset (including both weights and architectures) for FashionMNIST, CIFAR-10, and CIFAR-100 [25]. These datasets enable evaluation of ensemble methods without the need to train large numbers of models from scratch. Training details are provided in Section 3.1.

1. Problem statement

Neural Architecture Search methods have proven effective at discovering high-performing individual models. However, many predictive tasks benefit from ensemble strategies that aggregate multiple models rather than relying on a single architecture. The challenge lies in the fact that an optimal ensemble requires not merely a collection of strong models, but a set of architectures that actively complement one another. This motivates our formulation of Neural Ensemble Search as a unified optimization problem, where architectural selection and ensemble composition are jointly optimized. We begin by reviewing the standard NAS framework, then extend it to the ensemble setting.

1.1 Neural Architecture Search

We define the core components of the search space as:

$$\mathcal{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_N\} \subset \mathbb{R}^d, \quad \mathcal{O} = \{\mathbf{o}_1, \dots, \mathbf{o}_K\}, \mathbf{o}_i : \mathbb{R}^d \rightarrow \mathbb{R}^d,$$

where $\mathcal{V}$ is the ordered set of node feature-vectors (implying a topological order), and $\mathcal{O}$ is a finite set of elementwise operations (e.g., convolutions, poolings).

Using these components, edges are defined as annotated directed relations, represented by tuples of the form $(\mathbf{v}_i \rightarrow \mathbf{v}_j, \mathbf{o})$. We then define the space of feasible architectures $\mathcal{A}$ as the set of all labeled directed acyclic graphs, where edges strictly follow the topological order to ensure acyclicity:

$$\mathcal{A} = \{(\mathcal{V}, E) \mid E \subseteq \{(\mathbf{v}_i \rightarrow \mathbf{v}_j, \mathbf{o}) \mid 1 \leq i < j \leq N, \mathbf{o} \in \mathcal{O}\}\}.$$

Here, the condition $i < j$ restricts edges to forward connections only, intrinsically guaranteeing that every architecture in $\mathcal{A}$ is a Directed Acyclic Graph (DAG).

Consequently, a specific architecture $\alpha \in \mathcal{A}$ can be represented as:

$$\alpha = (\mathcal{V}, \{(\mathbf{v}_i \rightarrow \mathbf{v}_j, \mathbf{o}_{i,j}) \mid (\mathbf{v}_i \rightarrow \mathbf{v}_j) \in \mathcal{E}_\alpha\}),$$

where $\mathcal{E}_\alpha$ is the set of active edges satisfying $1 \leq i < j \leq N$, and each edge is uniquely labeled by an operation $\mathbf{o}_{i,j} \in \mathcal{O}$.

Denote $\mathcal{L}_{train}$ and $\mathcal{L}_{val}$ as the training and validation losses, respectively. The NAS problem can then be formulated as the search for an optimal architecture α^* that minimizes $\mathcal{L}_{val}(\alpha^*, \omega^*)$, under the constraint that the weights are obtained by minimizing the training loss:

$$\omega^* = \arg \min_{\omega \in \mathcal{W}} \mathcal{L}_{train}(\alpha^*, \omega).$$

This can be expressed as the following optimization problem:

$$\begin{aligned} \min_{\alpha \in \mathcal{A}} \quad & \mathcal{L}_{val}(\alpha, \omega^*(\alpha)), \\ \text{s.t.} \quad & \omega^*(\alpha) = \arg \min_{\omega \in \mathcal{W}} \mathcal{L}_{train}(\alpha, \omega). \end{aligned} \tag{1.1}$$

The primary challenge in this optimization lies in the immense search space of possible architectures (e.g., in DARTS [26], it is approximately 10^{25}).

1.2 Neural Ensemble Search

The primary objective of NES is to find an optimal ensemble of neural networks with architectures constrained to the same architectural search space as that explored by NAS methods.

As before, we denote $\alpha \in \mathcal{A}$ as a network architecture and $\omega(\alpha)$ as its corresponding parameters. The output of this network on an input x is denoted by $f_\alpha(x, \omega(\alpha))$. Let $S \subset \mathcal{A}$ denote a subset of architectures. Then, the NES problem can be formally described as follows:

$$\begin{aligned} \min_S \quad & \mathcal{L}_{val} \left(\frac{1}{|S|} \sum_{\alpha \in S} f_\alpha(x, \omega^*(\alpha)) \right), \\ \text{s.t.} \quad & \forall \alpha \in S : \omega^*(\alpha) = \arg \min_{\omega(\alpha)} \mathcal{L}_{train}(f_\alpha(x, \omega(\alpha))). \end{aligned} \tag{1.2}$$

Thus, in addition to searching over a vast number of architectures, we now also need to find the optimal ensemble composition.

1.3 Theoretical Framework

In this section, we establish a theoretical relationship between the oracle accuracy of a model pair and their pairwise prediction disagreement. This decomposition provides a formal justification for explicitly optimizing diversity in neural ensemble search. The following results provide the theoretical foundation relating prediction disagreement to ensemble performance.

We introduce the following definitions for two classification models f_i and f_j operating on a C -class problem:

- $a_i = \mathbb{P}(f_i(X) = Y)$ and $a_j = \mathbb{P}(f_j(X) = Y)$ denote the individual validation accuracies.
- $\gamma_{ij} = \mathbb{P}(f_i(X) \neq f_j(X))$ denotes the prediction disagreement rate.
- $\delta_{ij} = \mathbb{P}(f_i(X) \neq Y, f_j(X) \neq Y, f_i(X) \neq f_j(X))$ denotes the *unproductive disagreement* (cases where both models err but predict different incorrect classes).
- $\text{Oracle}(f_i, f_j) = \mathbb{P}(f_i(X) = Y \vee f_j(X) = Y)$ denotes the oracle accuracy, i.e., the probability that at least one model is correct.

The following proposition is a standard identity in ensemble diversity analysis, see e.g. [27]. We restate it here in a form convenient for NES.

Proposition 1.1 (Oracle Accuracy Decomposition). *For any two classifiers f_i and f_j , the oracle accuracy satisfies the exact identity:*

$$\text{Oracle}(f_i, f_j) = \frac{a_i + a_j + \gamma_{ij} - \delta_{ij}}{2}. \quad (1.3)$$

Proof. Let us partition the joint prediction space into five mutually exclusive events for a random input (X, Y) :

$$\begin{aligned} p_1 &= \mathbb{P}(f_i(X) = Y, f_j(X) = Y), \\ p_2 &= \mathbb{P}(f_i(X) = Y, f_j(X) \neq Y), \\ p_3 &= \mathbb{P}(f_i(X) \neq Y, f_j(X) = Y), \\ p_4 &= \mathbb{P}(f_i(X) \neq Y, f_j(X) \neq Y, f_i(X) = f_j(X)), \\ p_5 &= \mathbb{P}(f_i(X) \neq Y, f_j(X) \neq Y, f_i(X) \neq f_j(X)). \end{aligned}$$

By definition, the individual accuracies, disagreement, and unproductive disagreement can be expressed as:

$$\begin{aligned} a_i &= p_1 + p_2, & a_j &= p_1 + p_3, \\ \gamma_{ij} &= p_2 + p_3 + p_5, & \delta_{ij} &= p_5. \end{aligned}$$

The oracle accuracy corresponds to the probability that at least one model is correct, which excludes only the joint error events p_4 and p_5 :

$$\text{Oracle}(f_i, f_j) = 1 - (p_4 + p_5) = p_1 + p_2 + p_3.$$

Now, consider the linear combination $a_i + a_j + \gamma_{ij} - \delta_{ij}$:

$$\begin{aligned} a_i + a_j + \gamma_{ij} - \delta_{ij} &= (p_1 + p_2) + (p_1 + p_3) + (p_2 + p_3 + p_5) - p_5 \\ &= 2p_1 + 2p_2 + 2p_3. \end{aligned}$$

Dividing by 2 yields the desired identity:

$$\text{Oracle}(f_i, f_j) = \frac{a_i + a_j + \gamma_{ij} - \delta_{ij}}{2}.$$

□

□

The next theorem provides an upper bound on the majority-vote error of an ensemble of binary classifiers. This result is closely related to the $\mathcal{C}$ -bound derived in the PAC-Bayesian framework by [28]. We adapt it to binary classifiers with individual accuracies above $1/2$ and express it in terms of average accuracy and average pairwise disagreement.

Theorem 1.1 (Bound on Majority-Vote Error for Binary Classifiers). *Let $f_1, \dots, f_K$ be binary classifiers with individual accuracies $a_i = \mathbb{P}(f_i(X) = Y) > \frac{1}{2}$. Let $\bar{a} = \frac{1}{K} \sum_{i=1}^K a_i$ denote the average accuracy, and $\bar{\gamma} = \frac{2}{K(K-1)} \sum_{i < j} \gamma_{ij}$ denote the average pairwise disagreement, where $\gamma_{ij} = \mathbb{P}(f_i(X) \neq f_j(X))$. The majority-vote error probability is $P_{\text{ens}} = \mathbb{P}\left(\frac{1}{K} \sum_{i=1}^K \mathbb{I}[f_i(X) \neq Y] < \frac{1}{2}\right)$.*

Provided that $1 - 2\frac{K-1}{K}\bar{\gamma} > 0$, the following upper bound holds:

$$P_{\text{ens}} \leq \frac{4\bar{a}(1 - \bar{a}) - 2\frac{K-1}{K}\bar{\gamma}}{1 - 2\frac{K-1}{K}\bar{\gamma}}. \quad (1.4)$$

Proof. Let $v_i = \mathbb{I}[f_i(X) = Y]$. By definition, $\mathbb{E}[v_i] = a_i$ and $\text{Var}(v_i) = a_i(1 - a_i)$. The joint probability that both classifiers f_i and f_j are correct is $\eta_{ij} = \mathbb{E}[v_i v_j] = \mathbb{P}(f_i(X) = Y, f_j(X) = Y)$.

In binary classification, the disagreement rate satisfies $\gamma_{ij} = \mathbb{P}(f_i(X) \neq f_j(X)) = a_i + a_j - 2\eta_{ij}$, which implies $\eta_{ij} = \frac{a_i + a_j - \gamma_{ij}}{2}$. The covariance between the correctness indicators is therefore:

$$\text{Cov}(v_i, v_j) = \eta_{ij} - a_i a_j = \frac{a_i + a_j - \gamma_{ij}}{2} - a_i a_j.$$

We now compute the variance of the average vote $\bar{v} = \frac{1}{K} \sum_{i=1}^K v_i$:

$$\begin{aligned} \text{Var}(\bar{v}) &= \frac{1}{K^2} \left(\sum_{i=1}^K \text{Var}(v_i) + \sum_{i \neq j} \text{Cov}(v_i, v_j) \right) \\ &= \frac{1}{K^2} \left(\sum_{i=1}^K (a_i - a_i^2) + \sum_{i < j} (a_i + a_j - \gamma_{ij} - 2a_i a_j) \right). \end{aligned}$$

Using the identities $\sum_{i < j} (a_i + a_j) = (K - 1) \sum_{i=1}^K a_i = K(K - 1)\bar{a}$ and $\sum_{i < j} 2a_i a_j = \left(\sum_{i=1}^K a_i \right)^2 - \sum_{i=1}^K a_i^2 = K^2 \bar{a}^2 - \sum_{i=1}^K a_i^2$, we obtain:

$$\begin{aligned} \text{Var}(\bar{v}) &= \frac{1}{K^2} \left(K\bar{a} - \sum_{i=1}^K a_i^2 + K(K - 1)\bar{a} - \sum_{i < j} \gamma_{ij} - K^2 \bar{a}^2 + \sum_{i=1}^K a_i^2 \right) \\ &= \bar{a}(1 - \bar{a}) - \frac{1}{K^2} \sum_{i < j} \gamma_{ij}. \end{aligned}$$

Substituting $\sum_{i < j} \gamma_{ij} = \frac{K(K-1)}{2} \bar{\gamma}$ yields the compact form:

$$\text{Var}(\bar{v}) = \bar{a}(1 - \bar{a}) - \frac{K - 1}{2K} \bar{\gamma}. \quad (1.5)$$

We apply Cantelli's inequality: for a random variable Z with mean μ and variance σ^2 , and any $t > 0$, $\mathbb{P}(Z - \mu \leq -t) \leq \frac{\sigma^2}{\sigma^2 + t^2}$. Setting $Z = \bar{v}$, $\mu = \bar{a}$, and $t = \bar{a} - \frac{1}{2} > 0$, we have:

$$P_{\text{ens}} = \mathbb{P}(\bar{v} < \frac{1}{2}) = \mathbb{P}(\bar{v} - \bar{a} \leq -t) \leq \frac{\text{Var}(\bar{v})}{\text{Var}(\bar{v}) + t^2}.$$

The denominator simplifies as follows:

$$\begin{aligned}\text{Var}(\bar{v}) + t^2 &= \bar{a}(1 - \bar{a}) - \frac{K-1}{2K}\bar{\gamma} + \left(\bar{a} - \frac{1}{2}\right)^2 \\ &= \bar{a} - \bar{a}^2 - \frac{K-1}{2K}\bar{\gamma} + \bar{a}^2 - \bar{a} + \frac{1}{4} \\ &= \frac{1}{4} - \frac{K-1}{2K}\bar{\gamma}.\end{aligned}$$

Substituting back and multiplying numerator and denominator by 4 gives the desired bound:

$$P_{\text{ens}} \leq \frac{4\bar{a}(1 - \bar{a}) - 2\frac{K-1}{K}\bar{\gamma}}{1 - 2\frac{K-1}{K}\bar{\gamma}}.$$

□

□

Corollary 1.1 (Monotonicity and Variance Constraint). *Under the assumptions of Theorem 1.1, for a fixed average accuracy $\bar{a} > \frac{1}{2}$, the upper bound on P_{ens} given in Equation (1.4) is strictly decreasing with respect to $\bar{\gamma}$. Furthermore, the non-negativity of the vote variance established in Equation (1.5) implies the structural constraint $\bar{\gamma} \leq \frac{2K}{K-1}\bar{a}(1 - \bar{a})$. As $\bar{\gamma}$ approaches this maximal admissible value, the theoretical error bound converges to zero, demonstrating that maximizing pairwise disagreement directly minimizes the worst-case majority-vote error.*

The following theorem presents a well-known decomposition of the Brier score for probabilistic ensembles, often referred to as the ambiguity decomposition. It is a direct extension of the results in [29] to the Brier score (a proper scoring rule). The identity shows that the ensemble's squared error is the average individual error minus a non-negative term that measures the diversity of the probability vectors.

Theorem 1.2 (Brier Score Decomposition for Ensembles). *Let C be the number of classes, $\mathbf{p}_i(\mathbf{x}) \in \mathbb{R}^C$ the probability output of the i -th model, and $\mathbf{y}(\mathbf{x})$ the one-hot encoding of the true class. Denote the ensemble prediction by $\bar{\mathbf{p}}(\mathbf{x}) = \frac{1}{K} \sum_{i=1}^K \mathbf{p}_i(\mathbf{x})$, and the individual and ensemble Brier scores by $\ell_i = \|\mathbf{y} - \mathbf{p}_i\|^2$ and $\bar{\ell} = \|\mathbf{y} - \bar{\mathbf{p}}\|^2$, respectively. Then*

$$\bar{\ell} = \frac{1}{K} \sum_{i=1}^K \ell_i - \frac{1}{2K^2} \sum_{i,j=1}^K \|\mathbf{p}_i - \mathbf{p}_j\|^2.$$

Proof. We start by expanding the average individual Brier score:

$$\begin{aligned}
\frac{1}{K} \sum_{i=1}^K \ell_i &= \frac{1}{K} \sum_{i=1}^K \|\mathbf{y} - \mathbf{p}_i\|^2 \\
&= \frac{1}{K} \sum_{i=1}^K (\|\mathbf{y}\|^2 - 2\mathbf{y}^\top \mathbf{p}_i + \|\mathbf{p}_i\|^2) \\
&= \|\mathbf{y}\|^2 - 2\mathbf{y}^\top \bar{\mathbf{p}} + \frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i\|^2.
\end{aligned}$$

Adding and subtracting $\|\bar{\mathbf{p}}\|^2$ gives

$$\frac{1}{K} \sum_{i=1}^K \ell_i = \|\mathbf{y} - \bar{\mathbf{p}}\|^2 + \frac{1}{K} \sum_{i=1}^K (\|\mathbf{p}_i\|^2 - \|\bar{\mathbf{p}}\|^2) = \bar{\ell} + \frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2.$$

Thus the ensemble Brier score decomposes as

$$\bar{\ell} = \frac{1}{K} \sum_{i=1}^K \ell_i - \frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2. \tag{1.6}$$

The second term is non-negative, showing that the ensemble error never exceeds the average individual error, and that a larger spread of the individual predictions reduces $\bar{\ell}$.

Now express the spread (ambiguity) in terms of pairwise distances. Using $\|\mathbf{p}_i - \bar{\mathbf{p}}\|^2 = \|\mathbf{p}_i\|^2 - 2\mathbf{p}_i^\top \bar{\mathbf{p}} + \|\bar{\mathbf{p}}\|^2$ and summing over i , we obtain

$$\frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2 = \frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i\|^2 - \|\bar{\mathbf{p}}\|^2.$$

Since $\|\bar{\mathbf{p}}\|^2 = \frac{1}{K^2} \sum_{i,j=1}^K \mathbf{p}_i^\top \mathbf{p}_j$, we have

$$\begin{aligned}
\frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2 &= \frac{1}{K} \sum_{i=1}^K \|\mathbf{p}_i\|^2 - \frac{1}{K^2} \sum_{i,j=1}^K \mathbf{p}_i^\top \mathbf{p}_j \\
&= \frac{1}{K^2} \sum_{i,j=1}^K (\|\mathbf{p}_i\|^2 - \mathbf{p}_i^\top \mathbf{p}_j).
\end{aligned}$$

The double sum is symmetric; replacing it by half the sum over all ordered pairs of the expression plus its counterpart with i and j swapped yields

$$\begin{aligned} \frac{1}{K^2} \sum_{i,j=1}^K (\|\mathbf{p}_i\|^2 - \mathbf{p}_i^\top \mathbf{p}_j) &= \frac{1}{2K^2} \sum_{i,j=1}^K [(\|\mathbf{p}_i\|^2 - \mathbf{p}_i^\top \mathbf{p}_j) + (\|\mathbf{p}_j\|^2 - \mathbf{p}_j^\top \mathbf{p}_i)] \\ &= \frac{1}{2K^2} \sum_{i,j=1}^K (\|\mathbf{p}_i\|^2 - 2\mathbf{p}_i^\top \mathbf{p}_j + \|\mathbf{p}_j\|^2) \\ &= \frac{1}{2K^2} \sum_{i,j=1}^K \|\mathbf{p}_i - \mathbf{p}_j\|^2. \end{aligned}$$

Substituting this into (1.6) gives the claimed identity.

□

□

Theorem 1.3 (Link between disagreement and distance in latent space). *Let $(X,Y) \sim P$, where $Y \in \{1, \dots, m\}$ is a finite set of classes, and let $\mathcal{A}$ be a space of neural network architectures. For each architecture $a \in \mathcal{A}$, let*

$$f_a : X \rightarrow Y$$

be the corresponding classifier.

Define the disagreement between two architectures by

$$D(a,b) = P(f_a(X) \neq f_b(X)).$$

Then there exist a Hilbert space $\mathcal{H}$ and a mapping

$$\Phi : \mathcal{A} \rightarrow \mathcal{H}$$

such that

$$D(a,b) = \frac{1}{2} \|\Phi(a) - \Phi(b)\|_{\mathcal{H}}^2.$$

Proof. Consider the Hilbert space

$$\mathcal{H} = L^2(X, P; \mathbb{R}^m).$$

Let $\{e_1, \dots, e_m\}$ be the standard orthonormal basis of $\mathbb{R}^m$.

Define the mapping

$$\Psi(a)(x) = e_{f_a(x)}.$$

Since $\|e_y\| = 1$ for every class label y , we have

$$\|\Psi(a)\|_{\mathcal{H}}^2 = \int_X \|\Psi(a)(x)\|^2 dP(x) = 1.$$

Hence $\Psi(a) \in \mathcal{H}$.

For any $a, b \in \mathcal{A}$ and any $x \in X$,

$$\langle \Psi(a)(x), \Psi(b)(x) \rangle = \langle e_{f_a(x)}, e_{f_b(x)} \rangle = \mathbf{1}[f_a(x) = f_b(x)].$$

Therefore,

$$\begin{aligned} \langle \Psi(a), \Psi(b) \rangle_{\mathcal{H}} &= \int_X \langle \Psi(a)(x), \Psi(b)(x) \rangle dP(x) \\ &= \int_X \mathbf{1}[f_a(x) = f_b(x)] dP(x) \\ &= P(f_a = f_b) \\ &= 1 - D(a, b). \end{aligned}$$

Using the polarization identity,

$$\begin{aligned} \|\Psi(a) - \Psi(b)\|_{\mathcal{H}}^2 &= \|\Psi(a)\|_{\mathcal{H}}^2 + \|\Psi(b)\|_{\mathcal{H}}^2 - 2\langle \Psi(a), \Psi(b) \rangle_{\mathcal{H}} \\ &= 1 + 1 - 2(1 - D(a, b)) \\ &= 2D(a, b). \end{aligned}$$

Hence

$$D(a, b) = \frac{1}{2} \|\Psi(a) - \Psi(b)\|_{\mathcal{H}}^2.$$

Setting $\Phi = \Psi$ completes the proof.

□

□

Implications for Neural Ensemble Search. Theorems 1.1 and 1.2 together provide a clear theoretical justification for explicitly promoting diversity in ensemble construction. Specifically, for a fixed level of individual model accuracy (or average Brier score), increasing the pairwise disagreement (or the spread of probability vectors) directly reduces an upper bound on the ensemble’s error (or the ensemble Brier score).

Theorem 1.3 establishes that the disagreement measure $D(a,b) = P(f_a(X) \neq f_b(X))$ admits an exact Hilbert-space representation, where pairwise disagreement is proportional to the squared distance between embedded architectures, and all embedded architectures lie on the unit sphere of the Hilbert space. This provides a theoretical justification for learning a surrogate embedding $S_\theta : \mathcal{A} \rightarrow \mathbb{R}^d$ using metric-learning objectives, as minimizing the triplet loss preserves the ordering induced by disagreement. The learned surrogate can be viewed as a finite-dimensional approximation of the Hilbert embedding Φ .

Together, these results establish a direct connection between latent-space geometry and ensemble quality. Increasing distances in the learned latent space corresponds to increasing prediction disagreement, which in turn is associated with improved diversity and tighter theoretical guarantees for ensemble performance. Therefore, searching for architectures that are simultaneously accurate and well separated in the learned latent space provides a principled strategy for constructing high-quality neural ensembles. This insight motivates the design of our surrogate-based diversity predictor presented in Section 2.2.

2. Surrogate-based ensemble search

In this work, we adapt the DARTS architecture space representation into a graph-structured form that can be processed by Graph Attention Networks (GATs) (see Section 2.1). In Section 2.2, we present the architecture of the surrogate functions and describe their operating principles. Section 2.3 then details the ensemble construction method based on these surrogate models, explaining how surrogate predictions are used to promote architectural diversity within the ensemble.

Together, these components enable the construction of ensembles that jointly consider predictive accuracy and architectural diversity.

2.1 Architecture Search Space

Following the conventions of [26], we instantiate the search space $\mathcal{A}$ using a cell-based approach. Each architecture $\alpha \in \mathcal{A}$ is composed of two functional units: a *normal cell* α_{norm} and a *reduction cell* α_{red} , such that $\alpha = (\alpha_{\text{norm}}, \alpha_{\text{red}})$. Both units are DAGs adhering to the formalisms established in Section 1.1.

Unlike the original DARTS, which relies on continuous relaxation, our method explores $\mathcal{A}$ via discrete random sampling. We consider cells composed of $n = 5$ nodes and $m = 10$ edges, with each node having exactly two incoming and two outgoing edges. This choice follows the general structural design of DARTS, while reducing the size of the search space and the computational cost of training the surrogate dataset. Given an operation set size $|\mathcal{O}| = 7$, the cardinality of the cell-level search space is:

$$|\mathcal{A}_{\text{cell}}| = \prod_{k=2}^{n-1} \binom{k}{2} \cdot |\mathcal{O}|^m \approx 10^9. \quad (2.1)$$

Consequently, the full architecture space $|\mathcal{A}| = |\mathcal{A}_{\text{cell}}|^2 \approx 10^{18}$ remains computationally intractable for exhaustive search, necessitating surrogate-guided exploration.

To train the dual-surrogate models, we construct a dataset $\mathcal{D}_{\text{train}}$ by evaluating N sampled architectures:

$$\mathcal{D}_{\text{train}} = \{(\boldsymbol{\alpha}_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N, \quad (2.2)$$

where $\mathbf{y}_i = f_{\boldsymbol{\alpha}_i}(\mathcal{X}_{\text{val}}, \boldsymbol{\omega}^*(\boldsymbol{\alpha}_i))$ is the vector of model predictions on a fixed validation subset $\mathcal{X}_{\text{val}}$, and acc_i is the corresponding validation accuracy. These observations serve as the ground truth for our surrogate functions.

2.2 Surrogate Function

To facilitate the construction of ensembles (Section 2.3), we predict both the performance and diversity of candidate architectures. Due to the enormous size of the architecture space, obtaining these characteristics via full training is infeasible. To address this, we employ surrogate models: given an architectural representation, they predict key properties of neural networks. We consider two surrogates:

$$f_{\text{acc}}^{\boldsymbol{\theta}} : \mathcal{A} \rightarrow \mathbb{R}, \quad \mathbf{f}_{\text{div}}^{\boldsymbol{\theta}} : \mathcal{A} \rightarrow \mathbb{R}^d.$$

Each architecture is represented as a directed acyclic graph (DAG), where nodes correspond to latent representations and edges to operations, similar to NAS-Bench-201 [30]. For surrogate modeling, we adopt an alternative graph representation, inspired by NAS-Bench-101 [31]: nodes represent operations and edges their corresponding latent embeddings. This conversion is performed as follows:

1. Each edge of the original graph is transformed into a node in the new graph, labeled by the operation present on that edge.
2. An oriented edge is drawn from operation $\mathbf{o}_i$ to $\mathbf{o}_j$ if in the original graph $\mathbf{o}_i$ acts from node $\mathbf{x}$ to $\mathbf{y}$ and $\mathbf{o}_j$ acts from $\mathbf{y}$ to $\mathbf{z}$.

Each operation is represented using a one-hot encoding. The surrogates themselves are Graph Attention Networks (GATs) with N sequential layers, residual connections, GraphNorm [32], a global pooling aggregation, and lightweight output heads (Figure 2.1).

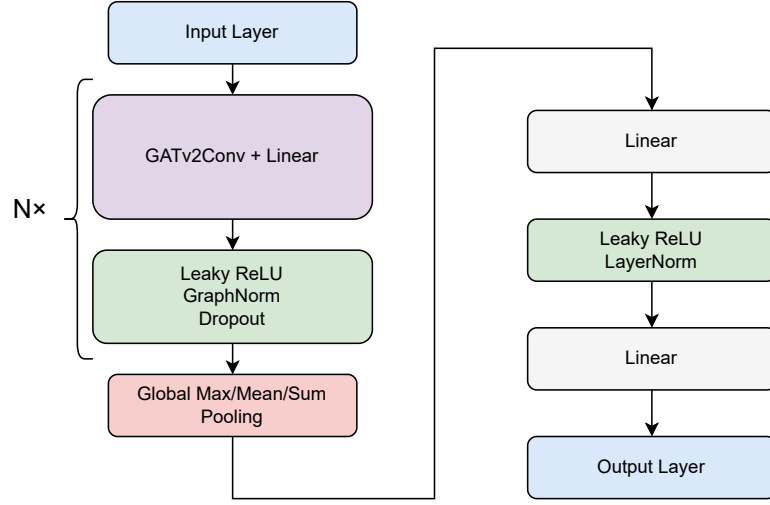


Figure 2.1 — Surrogate architecture: N GAT layers with residual connections and GraphNorm, global pooling, and output heads for accuracy and similarity embeddings.

The accuracy surrogate f_{acc} is trained via supervised regression to predict the performance of a given architecture. Given the training dataset $\mathcal{D}_{\text{train}} = \{(\boldsymbol{\alpha}_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N$, we optimize the parameters $\boldsymbol{\theta}_{\text{acc}}$ by minimizing the mean squared error (MSE) relative to the ground-truth validation accuracy:

$$\boldsymbol{\theta}_{\text{acc}}^* = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^N (f_{\text{acc}}^{\boldsymbol{\theta}}(\boldsymbol{\alpha}_i) - \text{acc}_i)^2, \quad (2.3)$$

In contrast, the diversity surrogate $\mathbf{f}_{\text{div}}^{\boldsymbol{\theta}}$ cannot be trained directly with supervised learning, since ground-truth latent embeddings are unavailable. Instead, we optimize it using a triplet loss on architecture embeddings.

Let us consider a set of I trained models $\{M_1, \dots, M_I\}$ and a fixed validation dataset of K examples. Denote the predictions of model M_i on the validation dataset by the vector

$$\mathbf{y}^{(i)} = (y_1^{(i)}, \dots, y_K^{(i)}).$$

Input: $\mathcal{D}_{train}$: training dataset of architectures $\{(\alpha_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N$;
D: diversity matrix where $\mathbf{D}_{jk} \in \{1, -1\}$ (positive/negative);
 m : triplet loss margin;
 η : learning rate;
 B : batch size.

Output: Optimized parameters θ_{div} for surrogate model $\mathbf{f}_{div}$

```

1 Initialize parameters  $\theta_{div}$ 
2 for epoch  $e \leftarrow 1$  to  $n$  do
3   for step  $t \leftarrow 1$  to  $N/B$  do
4     Sample a minibatch of anchor indices  $\mathcal{J} \subset \{1, \dots, N\}$ , where
        $|\mathcal{J}| = B$ 
5     for each  $j \in \mathcal{J}$  do
6       Identify candidate sets:  $\mathcal{P}_j = \{k \mid \mathbf{D}_{jk} = 1\}$  and
          $\mathcal{N}_j = \{k \mid \mathbf{D}_{jk} = -1\}$ 
7       Sample positive  $k_p \sim \text{Uniform}(\mathcal{P}_j)$  and negative
          $k_n \sim \text{Uniform}(\mathcal{N}_j)$ 
8       Compute embeddings:
9        $\mathbf{e}_a, \mathbf{e}_p, \mathbf{e}_n \leftarrow \mathbf{f}_{div}^{\theta_{div}}(\alpha_j), \mathbf{f}_{div}^{\theta_{div}}(\alpha_{k_p}), \mathbf{f}_{div}^{\theta_{div}}(\alpha_{k_n})$ 
10      Compute triplet loss for instance  $j$ :
11       $\ell_j \leftarrow \max(0, \|\mathbf{e}_a - \mathbf{e}_p\|_2^2 - \|\mathbf{e}_a - \mathbf{e}_n\|_2^2 + m)$ 
12    end
13    Compute batch loss:  $\mathcal{L} = \frac{1}{B} \sum_{j \in \mathcal{J}} \ell_j$ 
14    Update parameters:  $\theta_{div} \leftarrow \theta_{div} - \eta \nabla_{\theta_{div}} \mathcal{L}$ 
15  end
16 end
17 return  $\theta_{div}$ 

```

Figure 2.2 — Training procedure for the diversity surrogate function.

We construct a diversity matrix $\mathbf{C} \in [0, 1]^{I \times I}$, where each entry

$$c_{ij} = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(y_k^{(i)} = y_k^{(j)})$$

measures the fraction of matching predictions between models M_i and M_j .

To facilitate triplet-loss training, the diversity matrix $\mathbf{C}$ is discretized into $\mathbf{D} \in \{-1, 0, 1\}^{I \times I}$ according to upper and lower quantile thresholds q_p and q_n :

$$\mathbf{D}_{ij} = \begin{cases} 1, & \text{if } c_{ij} > q_p, \\ -1, & \text{if } c_{ij} < q_n, \\ 0, & \text{otherwise.} \end{cases}$$

The training procedure for the diversity surrogate $\mathbf{f}_{\text{div}}$ is detailed in Fig. 2.2. For each training epoch, we sample a batch of triplets $(\alpha_a, \alpha_p, \alpha_n)$ where the relationships are defined by the discretized diversity matrix $\mathbf{D}$. Specifically, for a given anchor architecture α_a , the positive α_p and negative α_n examples are sampled uniformly from the precomputed sets:

$$\mathcal{P}_a = \{k \mid \mathbf{D}_{ak} = 1\}, \quad \mathcal{N}_a = \{k \mid \mathbf{D}_{ak} = -1\}. \quad (2.4)$$

The architectures are subsequently mapped into a d -dimensional latent space via the surrogate function $\mathbf{f}_{\text{div}}$ with parameters θ_{div} :

$$\mathbf{e}_a = \mathbf{f}_{\text{div}}^{\theta_{\text{div}}}(\alpha_a), \quad \mathbf{e}_p = \mathbf{f}_{\text{div}}^{\theta_{\text{div}}}(\alpha_p), \quad \mathbf{e}_n = \mathbf{f}_{\text{div}}^{\theta_{\text{div}}}(\alpha_n). \quad (2.5)$$

The parameters θ_{div} are optimized by minimizing the triplet loss objective:

$$\mathcal{L}_{\text{div}} = \sum_{(a,p,n)} \max \left(\|\mathbf{e}_a - \mathbf{e}_p\|_2^2 - \|\mathbf{e}_a - \mathbf{e}_n\|_2^2 + m, 0 \right), \quad (2.6)$$

where $m > 0$ is a margin hyperparameter that enforces a minimum separation between dissimilar architectures in the embedding space. This optimization scheme encourages the surrogate to learn a manifold where Euclidean distances reflect architectural diversity.

The ground-truth matrix $\mathbf{D}$ used to guide the triplet sampling process is constructed by thresholding pairwise diversity metrics calculated on model outputs. In this work, we primarily utilize the fraction of identical predictions as the diversity measure, although the proposed framework is inherently compatible with other metrics, such as the Jensen-Shannon divergence, Hellinger distance, or the correlation of prediction vectors $\mathbf{y}_i$. These metrics yield an $I \times I$ diversity

Input: K : target ensemble size;
 N : number of initial candidates;
 λ : accuracy threshold;
 $f_{\text{acc}}, \mathbf{f}_{\text{div}}$: trained surrogate models.
Output: $\mathcal{A}_{\text{best}}$: selected ensemble of K architectures.

// Phase 1: Candidate Pool Construction

- 1 $\mathcal{A}_{\text{cand}} \leftarrow \text{RandomSample}(N)$
- 2 $\mathcal{P} \leftarrow \{(\alpha, a, \mathbf{e}) \mid \alpha \in \mathcal{A}_{\text{cand}}, a = f_{\text{acc}}(\alpha), \mathbf{e} = \mathbf{f}_{\text{div}}(\alpha), a \geq \lambda\}$

// Phase 2: Diversity-driven Greedy Selection

- 3 Select triplet with highest predicted accuracy:
 $(\alpha^*, a^*, \mathbf{e}^*) = \arg \max_{(\alpha, a, \mathbf{e}) \in \mathcal{P}} a$
- 4 $\mathcal{A}_{\text{best}} \leftarrow \{\alpha^*\}; \quad \mathcal{E}_{\text{sel}} \leftarrow \{\mathbf{e}^*\}; \quad \mathcal{P} \leftarrow \mathcal{P} \setminus \{(\alpha^*, a^*, \mathbf{e}^*)\}$
- 5 **while** $|\mathcal{A}_{\text{best}}| < K$ **and** $\mathcal{P} \neq \emptyset$ **do**
- 6 For each candidate $(\alpha_i, a_i, \mathbf{e}_i) \in \mathcal{P}$, compute
 $d_i = \frac{1}{|\mathcal{E}_{\text{sel}}|} \sum_{\mathbf{e}_{\text{sel}} \in \mathcal{E}_{\text{sel}}} \|\mathbf{e}_i - \mathbf{e}_{\text{sel}}\|_2$
- 7 Select triplet with maximum diversity:
 $(\alpha_{i^*}, a_{i^*}, \mathbf{e}_{i^*}) = \arg \max_{(\alpha_i, a_i, \mathbf{e}_i) \in \mathcal{P}} d_i$
- 8 Update ensemble: $\mathcal{A}_{\text{best}} \leftarrow \mathcal{A}_{\text{best}} \cup \{\alpha_{i^*}\}; \quad \mathcal{E}_{\text{sel}} \leftarrow \mathcal{E}_{\text{sel}} \cup \{\mathbf{e}_{i^*}\}$
- 9 $\mathcal{P} \leftarrow \mathcal{P} \setminus \{(\alpha_{i^*}, a_{i^*}, \mathbf{e}_{i^*})\}$
- 10 **end**
- 11 **return** $\mathcal{A}_{\text{best}}$

Figure 2.3 — Surrogate-Assisted Ensemble Construction.

matrix which is subsequently discretized to facilitate stable triplet sampling and the formation of a structured latent space.

2.3 Ensemble Construction

Once the surrogate models $f_{\text{acc}}^{\theta_{\text{acc}}}$ and $\mathbf{f}_{\text{div}}^{\theta_{\text{div}}}$ are trained, they enable an efficient search for the optimal ensemble without requiring the prohibitive computational cost of training intermediate candidate architectures. The proposed ensemble construction (Fig 2.3) proceeds in two distinct phases: candidate pool filtering and diversity-driven greedy selection.

In the first phase, we generate a large set of random candidate architectures $\mathcal{A}_{\text{cand}}$ via discrete sampling from the search space. We then use the dual-surrogate

framework to predict the accuracy s and the latent diversity embedding $\mathbf{e}$ for each candidate. To ensure the base quality of the ensemble, we apply an accuracy threshold α , forming a pruned candidate pool $\mathcal{P}$ that contains only high-performing models.

The second phase employs a greedy forward-selection strategy to maximize architectural diversity. Starting with the most accurate model as an anchor, the algorithm iteratively selects subsequent architectures that exhibit the maximum average Euclidean distance from the currently selected set in the latent space. By leveraging the geometric organization of the latent space provided by $\mathbf{f}_{\text{div}}$, this selection process explicitly favors architectures with dissimilar prediction patterns. This approach effectively identifies a diverse set of high-performing models, balancing individual predictive power with collective ensemble robustness.

3. Computational Experiment

In this section we present the experimental results as well as the metrics used for comparison. In Section 3.1, we describe the dataset collection procedure. Section 3.2 details the training setup for the surrogate functions. In Section 3.3, we present ablation studies analyzing the contribution of each surrogate function and the number of trained models required for surrogate learning. In Section 3.4, we measure how effectively the diversity function separates dissimilar architectures while attracting similar ones. Finally, in Sections 3.5, 3.6, 3.7, we compare our proposed method against DeepEnsemble [13] and Random Search [33] for FashionMNIST, CIFAR10 and CIFAR100, respectively.

3.1 Construction of the Training Dataset for Surrogate Models

The models in our dataset are trained according to the process described in Fig. 3.1. Training follows [26] with minor modifications; the main hyperparameters are summarized in Table 1.

Table 1 — Training Hyperparameters and Performance per Dataset

Dataset	Num. Cells	Initial Width	Num. Epochs	Avg. Acc. (%)	Avg. Top-1 Agreement
FashionMNIST	3	16	125	89.6 $\pm$ 0.5	0.900 $\pm$ 0.0
CIFAR-10	8	16	200	75.8 $\pm$ 0.6	0.693 $\pm$ 0.0
CIFAR-100	8	16	200	37.6 $\pm$ 1.1	0.324 $\pm$ 0.0

It is important to note that the architectures used to construct the surrogate training dataset are intentionally trained in a reduced configuration (in terms of depth and width) to limit computational cost. Moreover, these models are not trained to full convergence, but only for a fixed number of epochs sufficient to obtain reliable relative performance estimates.

In contrast, the final ensemble models are trained using a larger configuration, with increased number of cells and channel width. This design

allows the surrogate functions, learned on lightweight and partially trained models, to effectively guide the selection of higher-capacity architectures during ensemble construction, significantly reducing the overall computational cost.

Input: $\mathcal{A}_{DARTS}$: architectures available through DARTS;
 $\mathcal{X}$: dataset for train and validate architectures;
 N : number of architectures.
Output: Model dataset $\mathcal{D} = \{(\alpha_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N$.

- 1 Split dataset $\mathcal{X}$ into training and validation subsets: $\mathcal{X}_{\text{train}}, \mathcal{X}_{\text{val}}$ with a 20%/80% ratio
- 2 **for** $i \leftarrow 1$ **to** N **do**
- 3 Sample architecture $\alpha_i \sim \text{Uniform}(\mathcal{A}_{DARTS})$
- 4 Train model $f_{\alpha_i}(\mathcal{X}_{\text{train}}, \omega(\alpha_i))$
- 5 Evaluate f_{α_i} on $\mathcal{X}_{\text{val}}$
- 6 prediction vector $\mathbf{y}_i = f_{\alpha_i}(\mathcal{X}_{\text{val}}, \omega^*(\alpha_i))$
- 7 validation accuracy
- 8 $\text{acc}_i = \frac{1}{|\mathcal{X}_{\text{val}}|} \sum_{(x,y) \in \mathcal{X}_{\text{val}}} \mathbb{I}(\arg \max_c f_{\alpha_i}(x; \omega^*(\alpha_i)) = y)$
- 9 $\mathcal{D} \leftarrow \mathcal{D} \cup (\alpha_i, \mathbf{y}_i, \text{acc}_i)$
- 9 **end**
- 10 **return** $\mathcal{D}$

Figure 3.1 — Construction of the Model Evaluation Dataset

To improve efficiency, we employ an iterative surrogate-assisted refinement: we first train the accuracy surrogate on $N_1 = 1000$ randomly sampled architectures, then restrict the search to the top 10% candidates and sample an additional $N_2 = 2000$ architectures from this reduced space.

Constructing the surrogate dataset remains the main computational bottleneck. In our experiments, training approximately 3,000 architectures for each dataset, CIFAR-10 and CIFAR-100, required about three days per dataset on four NVIDIA A100 GPUs (80 GB each).

3.2 Training of the Surrogate Functions

In our experiment, each cell contains $n = 5$ nodes, with two outgoing edges per node assigned random operations from $\mathcal{O}$, yielding $m = 10$ edges per cell. The operation set $\mathcal{O}$ includes:

- Separable convolutions (3×3 , 5×5)
- Dilated separable convolutions (3×3 , 5×5)
- 3×3 max pooling
- 3×3 average pooling
- Skip connection

We employed a graph attention network (GAT) with four convolutional layers and two fully connected layers at the output, i.e. $N = 4$ (see Figure 2.1). The training of the surrogate functions used a cosine annealing learning rate scheduler.

Full models are constructed by stacking normal and reduction cells in a 2:1 ratio, starting from a predefined number of initial channels. Each model combines one normal and one reduction cell sampled independently from $\mathcal{A}$.

3.3 Surrogate Ablations

The main computational bottleneck of the proposed framework is the need to construct a large dataset of trained architectures. In this section, we analyze how the performance of the surrogate models depends on the training set size. All experiments are conducted on CIFAR-100 as the most challenging dataset.

We begin with the accuracy surrogate, which is used to filter candidate architectures based on their predicted performance. We hold out 600 architectures and evaluate how well the surrogate retrieves high-performing models using Spearman correlation and Recall@K.

The results are presented in Table 2. Both Spearman correlation and Recall@K improve as the number of training architectures increases, indicating that the surrogate becomes more effective at identifying strong models.

Table 2 — Surrogate performance as a function of training set size.

Type	Train Size	Spearman	R@10	R@50	R@100
Accuracy	0	$0.017_{\pm 0.051}$	$0.010_{\pm 0.030}$	$0.076_{\pm 0.017}$	$0.167_{\pm 0.021}$
	250	$0.106_{\pm 0.106}$	$0.040_{\pm 0.049}$	$0.130_{\pm 0.039}$	$0.228_{\pm 0.052}$
	500	$0.220_{\pm 0.058}$	$0.090_{\pm 0.083}$	$0.196_{\pm 0.064}$	$0.291_{\pm 0.034}$
	750	$0.218_{\pm 0.079}$	$0.120_{\pm 0.125}$	$0.204_{\pm 0.061}$	$0.291_{\pm 0.046}$
	1000	$0.368_{\pm 0.050}$	$0.120_{\pm 0.087}$	$0.264_{\pm 0.073}$	$0.330_{\pm 0.025}$
	1500	$0.535_{\pm 0.060}$	$0.170_{\pm 0.135}$	$0.294_{\pm 0.073}$	$0.366_{\pm 0.032}$
	2000	$0.675_{\pm 0.021}$	$0.200_{\pm 0.089}$	$0.324_{\pm 0.032}$	$0.430_{\pm 0.031}$
Diversity	0	$-0.001_{\pm 0.029}$	$0.019_{\pm 0.003}$	$0.092_{\pm 0.005}$	$0.179_{\pm 0.006}$
	250	$-0.490_{\pm 0.042}$	$0.029_{\pm 0.004}$	$0.131_{\pm 0.008}$	$0.249_{\pm 0.011}$
	500	$-0.545_{\pm 0.035}$	$0.031_{\pm 0.003}$	$0.136_{\pm 0.005}$	$0.261_{\pm 0.008}$
	750	$-0.574_{\pm 0.025}$	$0.029_{\pm 0.003}$	$0.135_{\pm 0.006}$	$0.259_{\pm 0.010}$
	1000	$-0.602_{\pm 0.025}$	$0.031_{\pm 0.003}$	$0.142_{\pm 0.007}$	$0.271_{\pm 0.012}$
	1500	$-0.639_{\pm 0.022}$	$0.032_{\pm 0.003}$	$0.147_{\pm 0.005}$	$0.281_{\pm 0.007}$
	2000	$-0.637_{\pm 0.025}$	$0.032_{\pm 0.004}$	$0.145_{\pm 0.007}$	$0.278_{\pm 0.008}$

The diversity surrogate is the core component of our method. To evaluate it, we again hold out 600 architectures. We measure (i) Spearman correlation between distances in the latent space and prediction similarity, and (ii) Recall@K, which reflects how well similar models are placed close to each other in the embedding space.

For the diversity surrogate, the Spearman correlation is negative, as larger distances in the latent space correspond to lower similarity between model predictions.

We observe that the diversity surrogate rapidly learns a meaningful latent structure: both correlation and Recall@K improve significantly when increasing the number of training architectures up to approximately 500–1000 samples, after which the gains become marginal. This suggests that reliable diversity estimation can be achieved without requiring the full surrogate training dataset.

We additionally perform a component ablation to assess the contribution of each surrogate function. The results are presented in Table 3.

Table 3 — Component ablation on CIFAR-100.

Method	Top-1 Acc. (%)	Avg. Model Acc. (%)	Pred. Disagreement
Random	84.41 $\pm$ 0.09	79.57 $\pm$ 0.19	0.430 $\pm$ 0.004
Accuracy only	85.07 $\pm$ 0.10	80.54 $\pm$ 0.12	0.406 $\pm$ 0.002
Diversity only	84.77 $\pm$ 0.07	80.08 $\pm$ 0.21	0.419 $\pm$ 0.008
Accuracy + Diversity	85.17 $\pm$ 0.16	80.50 $\pm$ 0.12	0.411 $\pm$ 0.004

3.4 Diversity Surrogate Function Ablations

To evaluate the behavior of the diversity surrogate, we first analyze the relationship between prediction similarity and embedding distance. Specifically, we compute the fraction of identical (top-1) predictions for pairs of models on a held-out dataset that was not used during training of the diversity surrogate, and compare it with the distances between their latent embeddings.

Figure 3.2 exhibits a consistent negative correlation across all three datasets: models producing more similar predictions tend to have closer embeddings. This observation confirms that the learned representation captures functional similarity between neural architectures.

Next, we study how the size of the training dataset influences the quality of the learned diversity representation. We vary the number of architectures used to train the diversity surrogate and quantify the agreement between embedding distances and empirical model similarity using Spearman’s and Kendall’s rank correlation coefficients.

The results are presented in Figure 3.3. Both metrics demonstrate a steady improvement as the number of training architectures increases, indicating that larger training sets lead to more reliable diversity estimation. Notably, a substantial portion of the performance gain is achieved even when the diversity surrogate is trained on a relatively small dataset of 500 architectures. This suggests that meaningful benefits can be obtained without resorting to prohibitively large and expensive training sets. For reference, training the full dataset of approximately 3,000 architectures for CIFAR-10 and CIFAR-100 required about three days on four NVIDIA A100 GPUs with 80 GB of memory each.

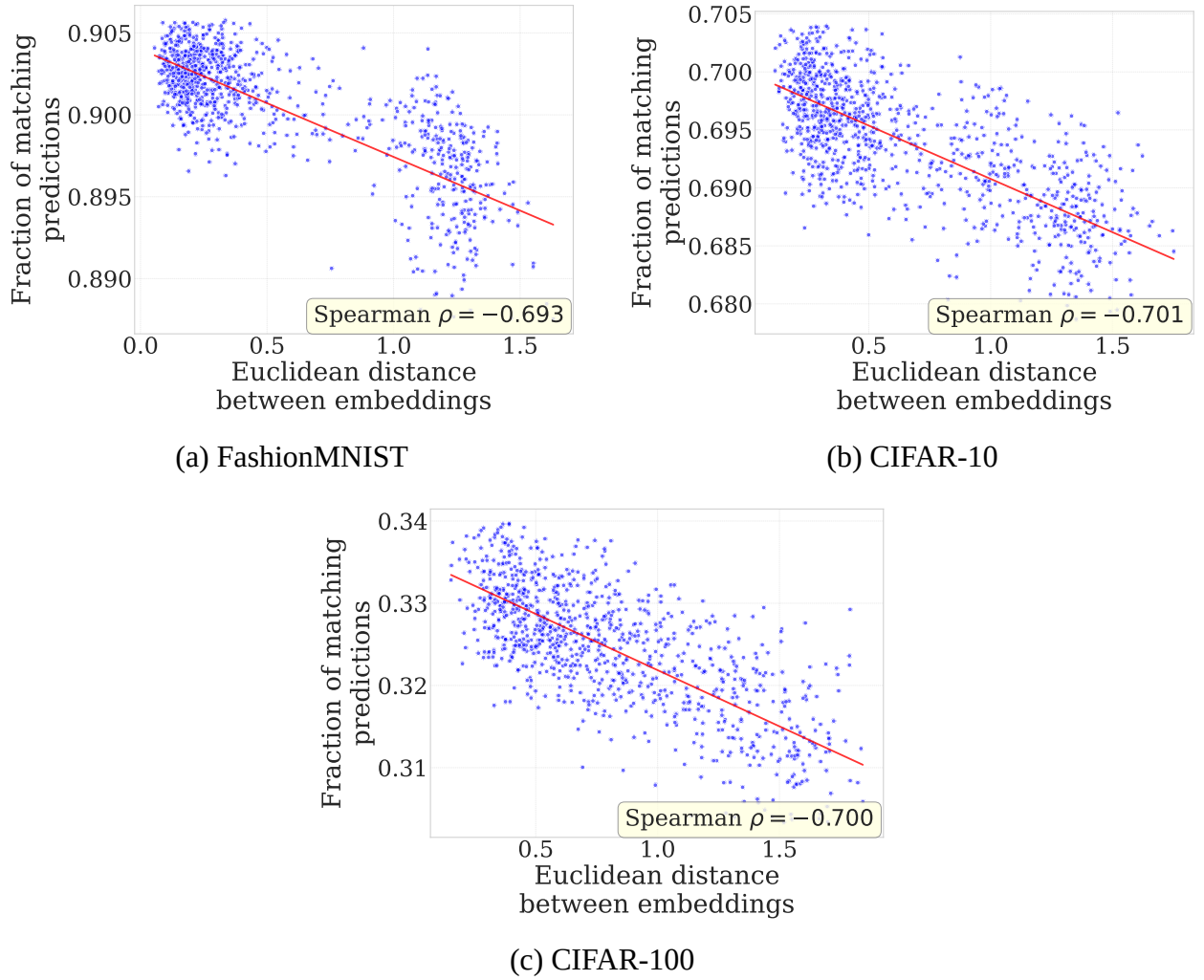


Figure 3.2 — Relationship between embedding distance and prediction similarity for pairs of models evaluated on a held-out dataset across different benchmarks.

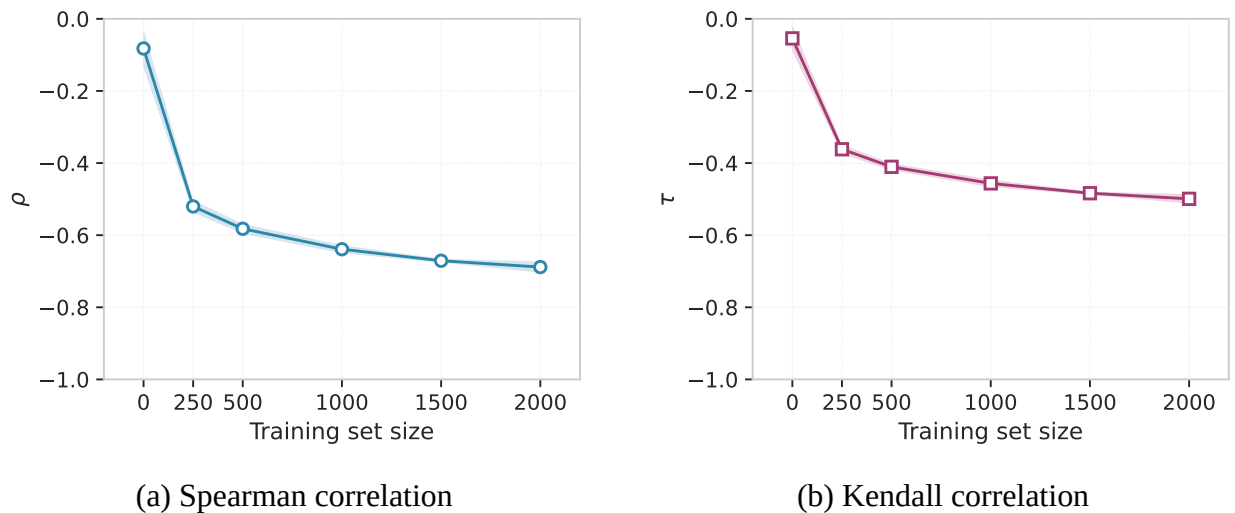


Figure 3.3 — Dependence of rank correlation between embedding distance and prediction similarity on the number of architectures used to train the diversity surrogate.

3.5 FashionMNIST

For constructing the diversity matrix, we used $q_p = 0.9$ and $q_n = 0.1$. In the diversity regime, the GAT surrogate has an output dimensionality of 16, employs 16 attention heads without dropout, and is trained for 50 epochs with a cosine-annealed learning rate from 5×10^{-4} to 1×10^{-5} . For accuracy prediction, the surrogate outputs a single scalar, uses 16 attention heads, and is trained for 125 epochs with a cosine-annealed learning rate from 5×10^{-3} to 1×10^{-4} . Both models are optimized with Adam on a 2400/600 train/validation split.

Subsequently, the models obtained via the surrogate functions were trained for 125 epochs using stochastic gradient descent (SGD) with a cosine-annealed learning rate, starting from 0.025 and decreasing to 0.001. Training was performed with a weight decay of 3×10^{-4} , an auxiliary loss weight of 0.4, a network composed of 3 cells with width 16, and a batch size of 96.

Table 4 reports the performance of five independently trained ensembles of three models each. The proposed Surrogate Ensemble achieves the highest Top-1 accuracy of 95.3%. An analysis of the relationship between individual model performance and ensemble efficacy yields several insights.

While the individual models identified by the DARTS-based DeepEns outperform those from Random Search on average (94.55% vs. 94.32%), both ensembles yield statistically equivalent Top-1 accuracies (95.02% and 95.01%, respectively). This phenomenon underscores the critical role of ensemble diversity. Specifically, Random Search leads across all evaluated diversity metrics, including Ambiguity, Normalized Disagreement, and Predictive Disagreement.

Furthermore, Random Search achieves the lowest Oracle NLL (0.191), suggesting that increased diversity compensates for lower individual model accuracy. These results provide empirical evidence that ensemble diversity remains a key performance driver even in primitive structured datasets like FashionMNIST.

Robustness analysis under adversarial attacks (Figure 3.4) shows comparable degradation across all methods. No single approach demonstrates a statistically significant advantage, suggesting that the intrinsic vulnerability of

Table 4 — FashionMNIST ensemble results.

Metric	Surrogate Ensemble	DeepEns	Random Search
Top-1 Accuracy (%)	95.3$\pm$0.1	95.02 $\pm$ 0.35	95.01 $\pm$ 0.19
Average Model Accuracy (%)	94.7$\pm$0.1	94.55 $\pm$ 0.43	94.32 $\pm$ 0.24
NLL	0.263$\pm$0.003	0.266 $\pm$ 0.006	0.265 $\pm$ 0.003
Oracle NLL	0.199 $\pm$ 0.010	0.206 $\pm$ 0.003	0.191$\pm$0.010
Brier Score	0.089$\pm$0.001	0.091 $\pm$ 0.006	0.092 $\pm$ 0.002
ECE	0.124 $\pm$ 0.002	0.121 $\pm$ 0.001	0.120$\pm$0.004
Ambiguity	0.0058 $\pm$ 0.0009	0.0047 $\pm$ 0.0009	0.0069$\pm$0.0013
Normalized Disagreement	0.065 $\pm$ 0.008	0.062 $\pm$ 0.008	0.077$\pm$0.009
Predictive Disagreement	0.130 $\pm$ 0.016	0.125 $\pm$ 0.016	0.154$\pm$0.018
FGSM AUC	0.058$\pm$0.0025	0.056 $\pm$ 0.0036	0.056 $\pm$ 0.0012
BIM AUC	0.032 $\pm$ 0.0037	0.033$\pm$0.0043	0.029 $\pm$ 0.0021
PGD AUC	0.023 $\pm$ 0.0018	0.024$\pm$0.0029	0.022 $\pm$ 0.0010

the search space on FashionMNIST outweighs the impact of ensemble selection strategies.

Finally, the effectiveness of the diversity surrogate is confirmed by the latent space analysis in Figure 3.2. The strong negative monotonic relationship (Spearman’s correlation $\rho = -0.69$) for 1,000 architecture pairs from a test set validates that $\mathbf{f}_{\text{div}}$ generalizes well to unseen models. However, the high density of pairs within a narrow embedding region confirms significant architectural redundancy on this dataset, which limits the potential for diversity-driven selection to further enhance ensemble performance.

3.6 CIFAR10

Diversity was measured using the overlap metric with thresholds $q_p = 0.9$ and $q_n = 0.1$. The GAT surrogates (16 dim/heads) were optimized for 20 and 100 epochs for the diversity and accuracy tasks, respectively, using cosine annealing. The selected architectures were subsequently trained for 600 epochs (20 cells, width 36) following the protocol described in Section 3.1, while keeping other hyperparameters unchanged.

The results in Table 5 are obtained from five independently trained ensembles of five models each. The proposed *Surrogate Ensemble* demonstrates superior predictive performance, achieving the highest Top-1 accuracy (97.80%) and the lowest NLL (0.208).

Table 5 — CIFAR10 ensemble results

Metric	Surrogate Ensemble	DeepEns	Random Search
Top-1 Accuracy (%)	97.80 ± 0.08	97.64 ± 0.09	97.56 ± 0.11
Average Model Accuracy (%)	96.73 ± 0.10	96.71 ± 0.14	96.48 ± 0.09
NLL	0.208 ± 0.003	0.210 ± 0.003	0.212 ± 0.002
Oracle NLL	0.156 ± 0.002	0.154 ± 0.004	0.158 ± 0.003
Brier Score	0.055 ± 0.001	0.056 ± 0.001	0.057 ± 0.001
ECE	0.136 ± 0.002	0.135 ± 0.002	0.136 ± 0.001
Ambiguity	0.011 ± 0.001	0.009 ± 0.001	0.011 ± 0.001
Normalized Disagreement	0.044 ± 0.001	0.046 ± 0.003	0.046 ± 0.001
Predictive Disagreement	0.087 ± 0.003	0.093 ± 0.007	0.092 ± 0.003
FGSM AUC	0.0569 ± 0.0012	0.0555 ± 0.0021	0.0559 ± 0.0012
BIM AUC	0.0333 ± 0.0032	0.0294 ± 0.0057	0.0309 ± 0.0034
PGD AUC	0.0273 ± 0.0015	0.0251 ± 0.0020	0.0256 ± 0.0018

Crucially, while our individual models share comparable accuracy with DeepEns ($\approx 96.7\%$), the ensemble benefits from enhanced diversity—evidenced by high Ambiguity (0.011) matching Random Search. This confirms that our dual-surrogate framework successfully balances high individual performance with the architectural heterogeneity required for complex datasets.

The benefit of this diversity extends to adversarial robustness (Figure 3.4), where our method maintains a consistent advantage under FGSM, BIM, and PGD attacks. Furthermore, latent space analysis (Figure 3.2) reveals a strong negative correlation (Spearman’s correlation $\rho = -0.7$) between embedding distance and prediction similarity. The wider dispersion in CIFAR-10’s embedding space allows the surrogate to effectively navigate a rich candidate pool, bridging the gap between the accuracy of DeepEns and the diversity of Random Search to achieve state-of-the-art performance.

3.7 CIFAR-100

Using the identical surrogate configuration as in Section 3.2, Table 6 reports results from five independently trained ensembles per method, each consisting of five models, demonstrating the effectiveness of the proposed approach on more complex datasets. Although DeepEns achieves slightly higher individual model accuracy (80.97% vs. 80.50%), our method prioritizes the selection of mutually complementary candidates, resulting in stronger collective performance and more consistent ensemble behavior. The Surrogate Ensemble attains a Top-1 Accuracy of 85.17%, outperforming both DeepEns (85.1%) and Random Search (84.41%).

Table 6 — CIFAR100 ensemble results.

Metric	Surrogate Ensemble	DeepEns	Random Search
Top-1 Accuracy (%)	85.17$\pm$0.16	85.10 $\pm$ 0.29	84.41 $\pm$ 0.09
Average Model Accuracy (%)	80.50 $\pm$ 0.12	80.97$\pm$0.41	79.57 $\pm$ 0.19
NLL	0.692 $\pm$ 0.007	0.682$\pm$0.010	0.718 $\pm$ 0.006
Oracle NLL	0.403$\pm$0.004	0.406 $\pm$ 0.007	0.418 $\pm$ 0.006
Brier Score	0.234 $\pm$ 0.002	0.232$\pm$0.004	0.244 $\pm$ 0.002
ECE	0.140 $\pm$ 0.004	0.133$\pm$0.003	0.141 $\pm$ 0.002
Ambiguity	0.047 $\pm$ 0.001	0.041 $\pm$ 0.002	0.048$\pm$0.002
Normalized Disagreement	0.206 $\pm$ 0.002	0.195 $\pm$ 0.005	0.215$\pm$0.002
Predictive Disagreement	0.411 $\pm$ 0.004	0.390 $\pm$ 0.010	0.430$\pm$0.004
FGSM AUC	0.0375$\pm$0.001	0.0368 $\pm$ 0.001	0.0358 $\pm$ 0.001
BIM AUC	0.0202$\pm$0.001	0.0181 $\pm$ 0.002	0.0196 $\pm$ 0.0007
PGD AUC	0.0172$\pm$0.0006	0.0158 $\pm$ 0.001	0.0167 $\pm$ 0.0005

Importantly, the results highlight that our method does not merely promote diversity, but ensures a principled and effective combination of complementary models. In contrast to Random Search, which increases diversity through stochastic selection and often includes sub-optimal architectures, our approach leverages the latent embedding structure-characterized by a strong correlation (Spearman’s $\rho = -0.7$) between embedding distance and prediction similarity (Figure 3.2)-to construct coherent ensembles. Regarding adversarial robustness (Figure 3.4), all methods exhibit comparable degradation, suggesting that on the complex CIFAR-100 manifold, the search space’s intrinsic vulnerability outweighs the benefits of ensemble selection strategies.

3.8 Adversarial Robustness Analysis

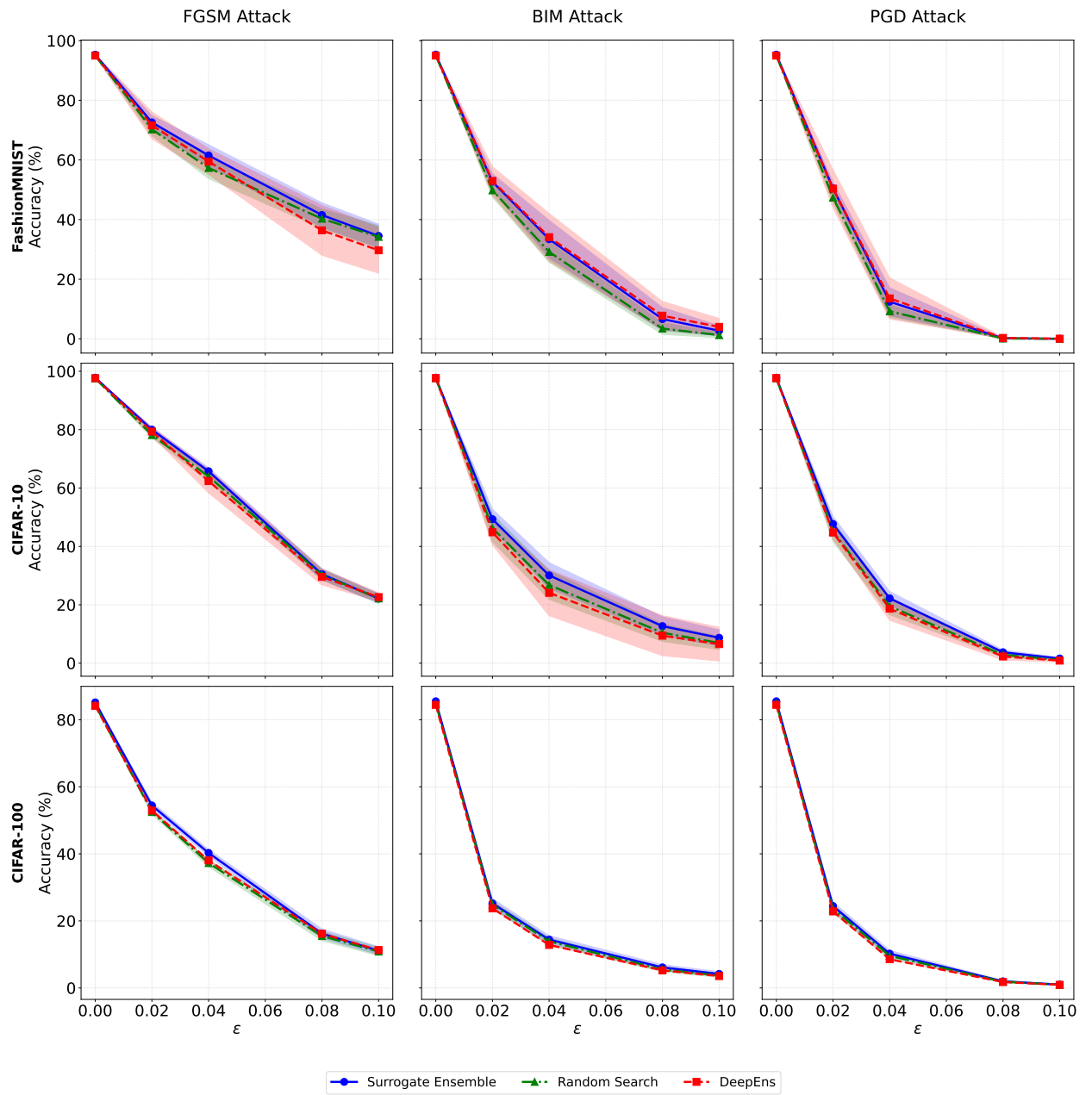


Figure 3.4 — Accuracy of the surrogate ensemble under FGSM, BIM, and PGD attacks across increasing ϵ for FashionMNIST, CIFAR-10 and CIFAR-100.

Conclusion

In this study, we demonstrated that surrogate diversity functions can be effectively learned through embedding distances in a latent space. Our experiments revealed consistently strong negative correlations between the Euclidean distance of model embeddings and the proportion of identical predictions across FashionMNIST, CIFAR-10, and CIFAR-100 (Spearman’s correlations ρ of -0.70 , -0.73 , and -0.71 , respectively). Building on these findings, we proposed an ensemble construction algorithm that simultaneously leverages accuracy and diversity surrogates to select high-performing, complementary models, demonstrating consistent improvements over standard baselines. Our results indicate that the efficacy of this approach is closely tied to the structural complexity of the dataset: while simple datasets favor ensembles composed of the most accurate individual models, more complex datasets allow diversity-driven selection to yield significant performance gains.

The primary limitation of the current approach is the substantial computational overhead required to train a large pool of base models for surrogate dataset generation. Future research should focus on mitigating these costs, perhaps through more efficient sampling or weight-sharing techniques.

Furthermore, we identify the investigation of the square root of Jensen-Shannon divergence as a promising alternative similarity metric. Unlike discrete response-proportion measures, a continuous metric based on Jensen-Shannon distance may provide a smoother optimization landscape, potentially refining diversity estimates. Additionally, situating these empirical findings within the formal framework of the bias-variance-covariance decomposition could offer deeper theoretical insights into the fundamental interplay between model accuracy and architectural diversity in ensemble performance.

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